PAVAN KUMAR KARIMI

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OBJECTIVE

Motivated and results-driven recent graduate with a strong foundation in software development and applications. Possesses excellent technical skills in programming, problem-solving, and data analysis, with hands-on experience in developing machine learning models and web applications. Adept at handling complex tasks, optimizing performance, and managing time effectively to meet deadlines. Eager to contribute to a dynamic team and apply knowledge in a real-world setting

SKILLS SUMMARY

- Languages: HTML, CSS, JavaScript, SQL, Python, C, Data Structures
- Libraries: Bootstrap, Reacts, NumPy, Pandas
- Tools & Platforms: GIT, Visual Studio Code, Jupiter Notebook, MYSQL, MS-Office
- Soft Skills: Problem-solving, Team Work, Willingness to Learn, Adaptability

EDUCATION

NRI Institute of Technology Vijayawada, India
Bachelor of Technology – **Information Technology** GPA: **8.7** *Jan 2018- May 2022*

Courses: Data Structures and Algorithms, AI, Machine Learning, Deep Learning, Natural Language Processing

WORK EXPERIENCES

NextHub Technologies PVT LTD May -2022 as Technical Trainer

- Technical trainer responsible for teaching technical concepts, guiding students, and delivering clear, practical training sessions. Ensures learners understand tools, technologies, and industry-ready skills

BIST Technologies Pvt. LTD Data Science (Virtual)

- **Project Overview:** Developed a deep Learning model to fuse textual and visual information from medical image modality for improved diagnostic accuracy.
- **Key Contributions**: Developed a multimodal deep learning model combining CNNs, RNNs, and attention mechanisms for accurate prediction from medical images and text. Optimized data preprocessing for both textual metadata and images to ensure consistency and reliability. Built a responsive frontend using HTML, CSS, Bootstrap, and React.js for user-friendly data input and result visualization
- **Performance Achievement**: Completed the internship with an assessment grade of A^+.

Blackbuck's Summer (Machine Learning)

- **Project Overview:** Built a machine learning pipeline to detect pneumonia from chest X-ray images, reducing diagnostic errors.
- **Key Contributions**: Developed and implemented a CNN model using TensorFlow and Keras, optimizing performance through advanced data preprocessing techniques like resizing, normalization, and augmentation. Fine-tuned hyper parameters to enhance model accuracy and efficiency. Built a responsive frontend using HTML, CSS, Bootstrap, and React.js for user-friendly data input and result visualization
- **Performance Achievement**: Completed the internship with an assessment grade of A^+.

PROJECTS

- **Task Management Application:** Built a full-stack web app using React.js, Flask, and MySQL to manage daily tasks with features like user authentication, task creation, and reminders. Focused on responsive design and user-friendly interface.

- **Pneumonia Detection Using Machine Learning**: Full-stack web application to detect pneumonia from chest X-ray images using Convolutional Neural Networks (CNN), achieving 95% accuracy. The backend was built with Python and integrated ML models with image preprocessing (resizing, normalization, augmentation). The frontend was developed using React.js for user-friendly X-ray image uploads and result display.

CERTIFICATIONS

- AI-ML Intern, **Blackbucks**
- AWS Cloud Internship, **Verzeo**
- Microsoft Azure Data Engineer Associate
- Introduction to Machine Learning, **Kaggle**
- The Joy of Computing Using Python, **NPTEL**
- Cloud Computing Certification, **NPTEL**
- Core Java Certification, **Udemy**
- Detection of Pneumonia in Chest X-Rays Using Convolutional Neural Networks," published in **ijisrt**, 2021.

ACHIEVEMENTS

- Achieved 150+ problem-solving milestones on platforms like LeetCode and Hacker Rank.
- Delivered impactful ML and AI solutions during internships, enhancing organizational workflows.

ROLES AND RESPONSIBILITIES

- Write clean Python 3.x scripts to read CSV / Excel files and load rows into SQLite with pandas + SQL Alchemy.
- Build small Airflow DAGs (or schedule with corn) to run daily file pickups and email success/failure alerts.
- Create pedantic models and simple data classes to validate incoming data before insert.
- Build and reuse React components every day to fetch, validate, and show data from REST APIs.
- Let users upload CSV/Excel/JSON files, parse them in the browser, and preview the rows before sending to the server.
- Create smart forms with live error messages using React-Hook-Form + Yup so users know instantly if something is wrong.
- Update old components to match new Figma or Storybook specs without breaking the current look, feel, or keyboard/screen-reader support.
- Add new fields, hide/show parts of the page, and write small custom hooks or memorized selectors whenever the business asks for more.
- Store frequently used data in memory or Indexed DB so the app talks to the network less and feels faster.
- Write modern ES6+ JavaScript and style components with styled-components or CSS-in-JS; keep class names short and consistent.
- Run React Profiler, Lighthouse, and Web-Vitals to find slow spots and make the app load and respond quicker

DECLARATION

- I hereby declare that all the above-mentioned information is true and correct to the best of my knowledge.

Signature

K. Pavan Kumar